AMD AI DEVMASTER HACKATHON 2026 · TRACK 2

Experience Asset

A Private Local Agent for Turning Lived Experience into Evidence-Bound, Reusable Knowledge

AMD AI DevMaster Hackathon 2026 — Track 2
 Document status: review draft; complete-product W7900 A0 and final demo pending
 Evidence snapshot: `main@d32730f`; PR #52 head `427d74f`
 Material snapshot ID: `EA-SUBMISSION-E9-20260806-R6`

EXECUTIVE SUMMARY

Personal experience contains decisions, constraints, evidence and lessons, but ordinary notes and one-shot AI writing flatten them into prose that is hard to verify, correct, resume or reuse.

Experience Asset is a private local Agent that turns one first-person experience into a versioned, evidence-bound asset. The user progresses through five visible artifacts, corrects facts before downstream reuse, and prepares audience-specific deliverables from one canonical version.

React and FastAPI provide the product surface. LangGraph maintains the durable multi-step workflow. A deterministic Runtime owns tool permissions, argument validation, phase-gated retrieval, privacy, approval, retries, persistence and verification. The model proposes structured actions but cannot authorize execution or disclosure. Core inference is designed for loopback vLLM, ROCm and Radeon PRO W7900.

2. Application Scenario, Users and Non-goals

The target user has a real experience, practice, professional insight or product exploration worth preserving, but finds it difficult to separate what happened from what was judged, inferred or left uncertain. The primary task is not “write me a post.” It is “help me build a correctable asset that I can inspect, continue and responsibly reuse.”

The engineering demo uses a public, anonymized office-chair trial, return and replacement experience. P0 excludes automatic publishing, platform growth strategy, traffic prediction, medical/legal/financial advice, unreviewed web research, raw multimodal understanding and multi-user collaboration.

The product does not claim that one person's experience is universal or that hallucinations are eliminated. It exposes sources, uncertainty, boundaries and permission decisions so unsupported output is easier to detect and block.

3. Product Workflow and Outputs

Milestone	Visible artifact	Core question
M1	Experience Brief	What happened, what is the source boundary, and what does the user want?
M2	Critical Moment Map	Which cues and turning points changed the experience?
M3	Decision and Evidence Ledger	What was decided, why, what action followed, and what evidence supports the result?
M4	Insight and Boundary Map	What was learned, what is the counterfactual, and where does the lesson stop applying?
M5	Route Readiness Matrix	Which audience routes are eligible, private and approved?

Each milestone is editable, saved, resumable and versioned. A material correction creates a new canonical identity and invalidates dependent evaluation, approval or export state. The canonical asset supports in-app preview plus consistent HTML, Markdown and JSON exports. Route selection never overrides deterministic eligibility, privacy or approval policy.

At PR #52 head `427d74f`, the integrated path contains production M5 LangGraph, restart-safe state, route/language identity handling, privacy resolution, exact-bound approval and the save→preview→export delivery surface. This is locally verified integration behavior; it is not yet complete-product W7900 A0.

4. Agent Architecture and Track 2 Capabilities

```
React UI
  -> FastAPI
  -> LangGraph + deterministic Runtime
       planning and checkpoints
       tool allowlist and argument validation
       phase-gated RAG and evidence identities
       privacy, approval, retries and terminal states
       SQLite versions and audit
  -> localhost OpenAI-compatible vLLM
  -> ROCm
  -> Radeon PRO W7900
```

The observable Agent loop is: inspect state, choose a bounded step, retrieve authorized evidence, propose a structured action, validate policy, execute an allowed effect, verify the result, then persist or stop safely.

This demonstrates the Track 2 capabilities of multi-step planning, local RAG, governed tool invocation, persistent multi-turn memory and permission/privacy control. The Agent application and vLLM remain in separate environments, and private services bind to `127.0.0.1` by default.

5. Grounding, Phase-gated RAG, Privacy and Approval

Material statements remain typed as source fact, user judgment, Agent inference or unresolved information. Schemas, source-span identities, deterministic recomputation and version invalidation constrain unsupported or stale output.

```
M1–M4: current source and prior versions of this asset only
  -> M4 Knowledge Card reaches explicit readiness
  -> summary-only cross-asset candidate discovery
  -> user confirms one relation
  -> detailed retrieval from that relation only
```

No match, declined relation and permission denial are normal outcomes. External output requires privacy review and approval bound to the exact canonical version, relation evidence, route, language, policy and renderer input. A material change makes the approval stale. The current route-draft path validates statement identities and route-bound evidence links. It does not yet claim complete text-level entailment between every editable sentence and its cited statement, so final wording remains subject to user or editor review.

6. Local Model Deployment and Radeon Optimization

The intended submitted inference path is local and OpenAI-compatible through vLLM on ROCm/W7900. The final application model remains an A0 evidence slot and must be recorded with exact name, revision, hash, precision, context window and runtime manifest.

Two bounded device evidence lines already exist:

- exact commit `60503eb` retains earlier M1/M2 product evidence on W7900 with

local Qwen2.5-14B-Instruct; it is not the complete M1–M5 product;

- PR #44, current head `69990dd`, packages a scoped Qwen3.5-9B BF16/8K W7900

Stage 1 service result. It does not select the submitted application model or prove complete-product gain.

In the frozen 8K Stage 1 comparison, P1 versus S0 produces a Stage1Score of `105.079709` versus `100.000000` (+5.08%), lowers p95 TTFT from `1710.910 ms` to `275.670 ms` (83.89%), lowers median E2EL from `34763.366 ms` to `33256.475 ms` (4.33%), and raises output throughput from `16.232` to `16.308 tokens/s` (0.47%). P1 reliability is 85% and must be shown with the result. All five confirmed Stage 1 profiles passed nine hard gates; the frozen evidence pack contains 526 artifacts.

PR #44 also records a bounded 20K service-usability PASS for S0/P1/P4/P3. The original strict business-semantics failures remain retained as advisories. This overlay does not change the frozen ranking, promote a product default or prove the complete application at 20K.

Optimization is correctness-first. A0 must freeze the complete application, model, workload, environment and quality/privacy gates. Only then may a Stage 2 candidate support a comparable claim using TTFT, output throughput, end-to-end task time, peak memory, completion rate and failure recovery. No complete-product performance improvement is claimed in this draft.

PR #53 head `4912fe7` defines an optional same-Agent comparison between A0/S0 and A1/P1 after the complete product is frozen. Stage 2 is `NOT_STARTED` and is not required for the submitted product path. This document reports no complete-Agent optimization gain.

7. Current Validation and Limitations

Scope	Verified result	Boundary
Historical M1/M2 product path	W7900 + local Qwen2.5-14B-Instruct at `60503eb`	Early product only
Current E5–E7 integration	PR #52 head `427d74f`; production M5 graph and delivery path integrated	Open PR; local deterministic client, no live model/W7900
Current check surface	Targeted backend regressions, 34 public QA tests, 27 Web files / 222 tests, Ruff/lint/type/build/contracts and both GitHub checks pass	Do not combine into a final cross-stack test count
Stage 1 serving	Qwen3.5-9B BF16/8K Stage 1 pack in PR #44 head `69990dd`; P1 p95 TTFT –83.89%, median E2EL –4.33%, output TPS +0.47%, reliability 85%	Service only, not application A0
20K overlay	S0/P1/P4/P3 service-usability PASS in PR #44 head `69990dd`	Strict semantic advisories retained; no product promotion
Stage 2 paired Agent validation	A0=S0 and A1=P1 protocol in PR #53 head `4912fe7`	Protocol frozen; `NOT_STARTED`; no result claimed

The remaining critical evidence is one exact frozen application commit, one representative end-to-end product run with the real local model on W7900, its exact model/runtime manifest, final test ledger, A0 decision, final screenshots and the public approximately four-minute demo. Stage 2 may be reported as passed, fallback or `NOT_RUN`; it must not delay an honest complete-product submission.

8. Reproduction, Team and References

The cloud workspace uses `/workspace/amd-track2-agent`, keeps the Agent and vLLM in separate environments, and treats GitHub as the source of truth for code. The final frozen commit will provide the exact commands for bootstrap, environment verification, model serving, application startup, tests and the public fixture.

```
cd /workspace
git clone https://github.com/kinnyuww/amd-devmaster-track2-agent amd-track2-agent/repo
cd amd-track2-agent/repo
bash scripts/bootstrap_workspace.sh
bash scripts/verify_remote_dev.sh
bash scripts/verify_rocm_env.sh
bash scripts/serve_vllm.sh
```

The final contribution statement will be generated from traceable Git/PR history rather than planned ownership.

References:

- Official event and Rules & Conditions: https://luma.com/amd-4dhi
- Official organizer repository: https://github.com/AMD-DEV-CONTEST/Radeon-hackathon-2026-07
- Team repository and integrated product boundary: https://github.com/kinnyuww/amd-devmaster-track2-agent and https://github.com/kinnyuww/amd-devmaster-track2-agent/pull/52
- Stage 1 W7900 evidence: https://github.com/kinnyuww/amd-devmaster-track2-agent/pull/44
- Demo and PPT plan: https://github.com/kinnyuww/amd-devmaster-track2-agent/pull/55

The final package must also cite the exact ROCm/vLLM documentation, model license/revision and public-fixture provenance used by the frozen run.

Open final slots: application commit, model/runtime/device manifest, complete W7900 flow, A0/Stage 2 decision, final test ledger, screenshots, demo URL, public artifact links and contribution statement.